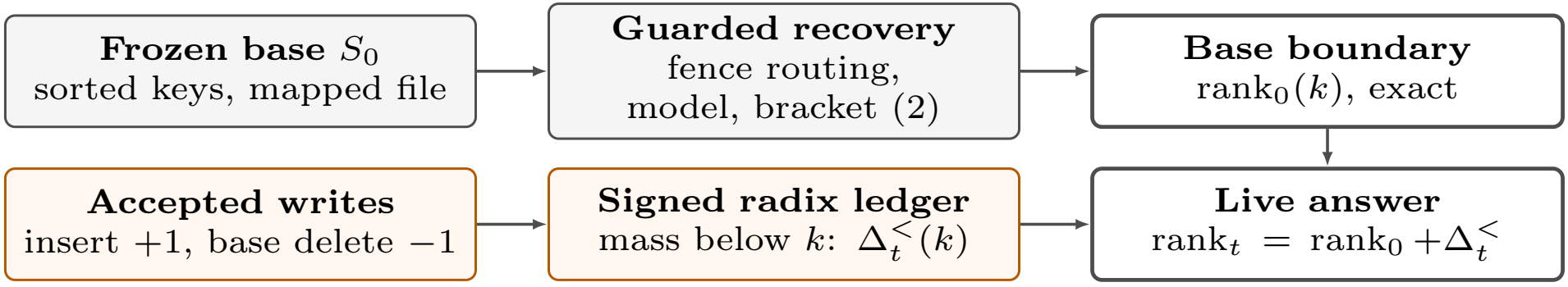
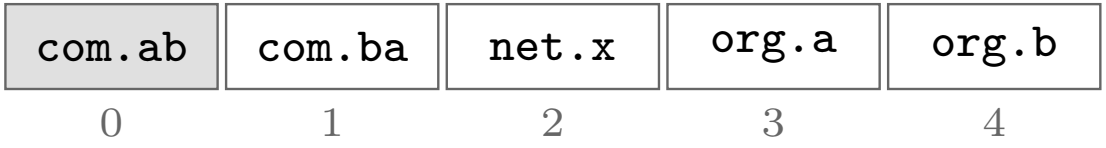
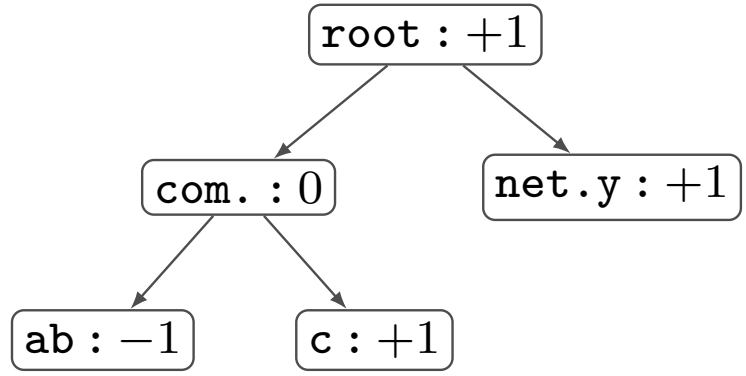
(a) Query path: the model is never touched by writes



(b) Frozen base S_0 (index = rank_0)



Signed ledger after writes



Writes: insert `com.c`, delete `com.ab`, insert `net.y`.
Stored key `org.a`: $\mathrm{rank}_0 = 3$; siblings before `o` carry $0 + 1$, so $\mathrm{rank}_t = 4$ and $p_t = p_0 + 1$ keeps p_0's error.
Absent boundary `com.b`: bracket finds $\mathrm{rank}_0 = 1$; inside `com.`, edge `ab` precedes `b`, so $\Delta^< = -1$ and $\mathrm{rank}_t = 0$.